

CS173: Discrete Structures

Problem Set 8

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1. An archaeologist on a recent expedition to the British Isles has discovered, for each positive natural number $n \in \mathbb{N}^+$, evidence of a manuscript written in a language called nglish. Each such manuscript is exactly n pages long and contains precisely $2n$ distinct words. Prove each manuscript contains a page with two distinct words on it.

Proof. Let $n \in \mathbb{N}^+$ be arbitrary, and consider an nglish manuscript with n pages and $2n$ distinct words. For each $i \in n$, let P_i denote the set of words on page i .

Since the manuscript contains $2n$ distinct words, $\left| \bigcup_{i=0}^{n-1} P_i \right| = 2n$.

Assume for contradiction that no page contains two distinct words, i.e., $(\forall i \in n)(|P_i| \leq 1)$. Observe:

$$2n = \left| \bigcup_{i=0}^{n-1} P_i \right| \leq \sum_{i=0}^{n-1} |P_i| \leq \sum_{i=0}^{n-1} 1 = n.$$

This gives $2n \leq n$, a contradiction since $n \in \mathbb{N}^+$.

Therefore $(\exists i \in n)(|P_i| \geq 2)$, i.e., a manuscript contains a page with two distinct words. $\square$

2. Show that there exist $a, b \in \mathbb{N}$ such that $256 < a < b$ and $7^a \equiv 7^b \pmod{512}$.

Proof. By the Euclidean Division Lemma,

$$(\forall a \in \mathbb{Z})(\forall b \in \mathbb{Z} \setminus \{0\})(\exists! q, r \in \mathbb{Z})(a = bq + r \wedge 0 \leq r < |b|).$$

Define $\% : \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) \rightarrow \mathbb{Z}$ by $x \% y := r$, where r is the unique integer guaranteed by the Euclidean Division Lemma satisfying $x = yq + r$ and $0 \leq r < |y|$ for some $q \in \mathbb{Z}$.

Let $A := \mathbb{N} \setminus \{k \in \mathbb{N} \mid k < 257\}$ and $B := \{k \in \mathbb{N} \mid k < 512\}$, so $|B| = 512$.

Define $f : A \rightarrow B$ by $(\forall x \in A)(f(x) := 7^x \% 512)$.

By the Corollary, $|A| = |\mathbb{N}|$. By the Theorem, an infinite set is larger than any finite set, so

$$|A| = |\mathbb{N}| > |B| = 512.$$

By the Pigeonhole Theorem,

$$|A| > |B| \Rightarrow (\forall f : A \rightarrow B)(\exists x, y \in A)(x \neq y \wedge f(x) = f(y)),$$

so $(\exists c, d \in A)(c \neq d \wedge f(c) = f(d))$, i.e., $7^c \% 512 = 7^d \% 512$.

By the Euclidean Division Lemma, $7^c = 512q_1 + r_1$ and $7^d = 512q_2 + r_2$ with $0 \leq r_1, r_2 < 512$. Since $f(c) = f(d)$, we have $r_1 = r_2$, so subtracting gives $7^c - 7^d = 512(q_1 - q_2)$. Since $q_1 - q_2 \in \mathbb{Z}$, we have $512 \mid (7^c - 7^d)$, i.e., $7^c \equiv 7^d \pmod{512}$.

Let $a := \min(c, d)$ and $b := \max(c, d)$. Since $c \neq d$ we have $a < b$, and since $c, d \in A$ we have $a, b > 256$.

Therefore there exist $a, b \in \mathbb{N}$ such that $256 < a < b$ and $7^a \equiv 7^b \pmod{512}$. $\square$

3. Consider the set $\mathcal{S} := \{3, 4, 7, 8, 9, 10, 12, 15, 18, 19, 27, 28\}$. Show that, in any subset $\mathcal{X} \subseteq \mathcal{S}$ of cardinality $|\mathcal{X}| \geq 9$, there exist three distinct elements $x_1, x_2, x_3 \in \mathcal{X}$ such that $x_1 + x_2 + x_3 = 40$.

Proof. Partition $\mathcal{S}$ into four triples:

$$G_1 = \{3, 10, 27\}, \quad G_2 = \{9, 12, 19\}, \quad G_3 = \{4, 8, 28\}, \quad G_4 = \{7, 15, 18\},$$

where $3 + 10 + 27 = 9 + 12 + 19 = 4 + 8 + 28 = 7 + 15 + 18 = 40$, the G_i are pairwise disjoint, and $G_1 \cup G_2 \cup G_3 \cup G_4 = \mathcal{S}$.

Let $\mathcal{X} \subseteq \mathcal{S}$ with $|\mathcal{X}| \geq 9$. Suppose for contradiction that no three distinct elements of $\mathcal{X}$ sum to 40. If $G_i \subseteq \mathcal{X}$ for some i , then the three distinct elements of G_i are in $\mathcal{X}$ and sum to 40, contradicting our assumption. Hence $|\mathcal{X} \cap G_i| \leq 2$ for each i .

Since $\{G_1, G_2, G_3, G_4\}$ partitions $\mathcal{S}$ and $\mathcal{X} \subseteq \mathcal{S}$, the sets $\mathcal{X} \cap G_1, \dots, \mathcal{X} \cap G_4$ are pairwise disjoint with union $\mathcal{X}$. Observe,

$$|\mathcal{X}| = \sum_{i=1}^4 |\mathcal{X} \cap G_i| \leq 4 \times 2 = 8,$$

contradicting $|\mathcal{X}| \geq 9$.

Therefore our claim that $|\mathcal{X} \cap G_i| \leq 2$ for all i is false, so there exists some i with $|\mathcal{X} \cap G_i| = 3$, i.e., $G_i \subseteq \mathcal{X}$. The three distinct elements of G_i are then in $\mathcal{X}$ and sum to 40, giving $x_1, x_2, x_3 \in \mathcal{X}$ with $x_1 + x_2 + x_3 = 40$. $\square$

4. For this problem, you may assume that $\binom{n}{0} = \binom{n}{n} = 1$ for all $n \in \mathbb{N}$. Prove that, for every $n, k \in \mathbb{N}$ such that $k \leq n$, we have $\binom{n}{k} = \binom{n}{n-k}$.

Proof. Let $n, k \in \mathbb{N}$ with $k \leq n$. Define

$$S_k := \{B \mid B \subseteq n \wedge |B| = k\}, \quad S_{n-k} := \{B \mid B \subseteq n \wedge |B| = n - k\},$$

so that $\binom{n}{k} = |S_k|$ and $\binom{n}{n-k} = |S_{n-k}|$ by definition.

Define $f : S_k \rightarrow S_{n-k}$ by $(\forall B \in S_k)(f(B) := n \setminus B)$.

Well-defined. Let $B \in S_k$. Then $f(B) = n \setminus B \subseteq n$, and by the Theorem ($B \subseteq n \Rightarrow |n \setminus B| = |n| - |B|$), we have $|n \setminus B| = n - k$, so $f(B) \in S_{n-k}$.

Injective. Let $B_1, B_2 \in S_k$ with $f(B_1) = f(B_2)$. Then $n \setminus B_1 = n \setminus B_2$. Since $B_1, B_2 \subseteq n$, we have $B_1 = n \setminus (n \setminus B_1) = n \setminus (n \setminus B_2) = B_2$.

Surjective. Let $C \in S_{n-k}$. Then $n \setminus C \subseteq n$, and by the Theorem, $|n \setminus C| = |n| - |C| = n - (n - k) = k$, so $n \setminus C \in S_k$. Then $f(n \setminus C) = n \setminus (n \setminus C) = C$.

Since f is a bijection, $|S_k| = |S_{n-k}|$, i.e., $\binom{n}{k} = \binom{n}{n-k}$. □

5. We define a **finite binary string** to be a function $b : n \rightarrow \{0, 1\}$, where $n \in \mathbb{N}$ is the length of the string. We define an **infinite binary string** to be a function $b : \mathbb{N} \rightarrow \{0, 1\}$.

How many infinite binary strings are there that each contain one contiguous finite substring consisting only of 1s? Formalize your claim and justify it with a proof.

Claim: There are countably many infinite binary strings containing exactly one contiguous finite substring of all 1s, i.e., the set of such strings has cardinality $|\mathbb{N}|$.

Proof. Let $\mathcal{T}$ denote the set of all infinite binary strings $b : \mathbb{N} \rightarrow \{0, 1\}$ such that the set $\{n \in \mathbb{N} \mid b(n) = 1\}$ is a nonempty finite contiguous interval.

For $i, k \in \mathbb{N}$ with $i \leq k$, define $b_{i,k} : \mathbb{N} \rightarrow \{0, 1\}$ by

$$b_{i,k}(n) := \begin{cases} 0 & \text{if } n < i \\ 1 & \text{if } i \leq n \leq k \\ 0 & \text{if } n > k \end{cases}$$

Let $S := \{(i, k) \in \mathbb{N} \times \mathbb{N} \mid i \leq k\}$ and define $f : S \rightarrow \mathcal{T}$ by $f(i, k) := b_{i,k}$.

Injective. Let $(i_1, k_1), (i_2, k_2) \in S$ with $f(i_1, k_1) = f(i_2, k_2)$, and let $b = b_{i_1, k_1} = b_{i_2, k_2}$. Then $i_1 = \min\{n \in \mathbb{N} \mid b(n) = 1\} = i_2$ and $k_1 = \max\{n \in \mathbb{N} \mid b(n) = 1\} = k_2$.

Surjective. Let $b \in \mathcal{T}$. Since b has exactly one maximal block of 1s, let $i := \min\{n \in \mathbb{N} \mid b(n) = 1\}$ and $k := \max\{n \in \mathbb{N} \mid b(n) = 1\}$. Then $(i, k) \in S$ and $f(i, k) = b$.

Since f is a bijection, $|S| = |\mathcal{T}|$. The set $\{(0, k) \mid k \in \mathbb{N}\} \subseteq S$ is infinite, so S is infinite. Since $S \subseteq \mathbb{N} \times \mathbb{N}$ and $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$, by Theorem, so $|S| = |\mathbb{N}|$. Therefore $|\mathcal{T}| = |\mathbb{N}|$.

Including the all-zero string adds one element to our count, giving $|\mathcal{T}| + 1$ total strings. Since $\mathcal{T}$ is infinite, adding a single element does not change its cardinality, so the total count remains $|\mathbb{N}|$. $\square$